

## Recent cloud bursts show — India needs early warning systems

It is becoming increasingly **evident** that **policymakers** in the country **need** to do more to **shield** people, whether in the plains, mountains, or **coastal** areas, from the **vagaries** of the monsoon. This year, cloudbursts, **flash floods**, and **landslides** have **caused havoc** in parts of Northeast India, Uttarakhand, Himachal Pradesh, and, most recently, in Jammu and Kashmir. **Waterlogging** and **crumbling infrastructure** have not only **disrupted** life and caused traffic **congestion** in urban centres, including cities such as Delhi, Bengaluru, and Mumbai, they have also **taken a high toll** of lives. **The writing has been on the wall** for at least a **decade** — most parts of the country need a **drainage** upgrade. This urban planning **imperative** is the easier of the monsoon-related challenges. Increasingly, it's also becoming evident that timely warning can reduce the loss of lives and minimise damage during extreme weather events. **Anticipating** nature's vagaries **requires** accurate and timely data and effective communication channels. **Work** on building such **mechanisms** **has** just begun in the country.

**The loss** of more than 60 lives in Jammu and Kashmir to cloudbursts in less than a week **has exposed** the **frailties** of the country's climate information system. The complex **processes** that go into the making of cloudbursts **have exacerbated** the challenges. The **interaction** between monsoon winds and cold winds **combines** with the **idiosyncrasies** of Himalayan **topography** to create huge clouds. In a region historically **prone to** cloudbursts, the **effects** of **global warming** **seem** to have been **treacherous**. When the clouds can no longer hold the water, they **unleash** rain bombs over relatively small areas. **Meteorologists** define a cloudburst as rain falling at a rate of 100 mm or more per hour over an area of about 30 sq km. Such small **tracts** often lack rainfall-measuring instruments. The **effects** of the **downpour**, however, **are** not localised — they **trigger** landslides and flash floods and cause destruction downstream. **Theoretically**, it is not impossible to **forecast** rainfall over a very small area, but it requires a **dense** network of weather instruments and **computing** capabilities. The India Meteorological Department (IMD) is **reportedly stepping up** initiatives to fill information gaps as well as make maximum use of data.

The **task** of shielding people from the effects of extreme rainfall **is**, however, too **enormous** to be left to the IMD. **Saving** lives in **fragile ecosystems** **will** require **coordination** between the met office, scientists, planners, and local, state, and national-level authorities. **For starters**, detailed **maps** of cloudburst-prone zones that use historical data and satellite imagery **could** guide land-use plans. They could ensure construction and developmental activities are carefully regulated and climate-proofed in high-risk areas. It's an imperative that cannot be postponed.

[Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

## Vocabulary

1. **Cloud burst** (noun) – torrential downpour, rain bomb, sudden deluge, cloudburst storm, extreme rainfall बादल फटना
2. **Evident** (adjective) – clear, obvious, apparent, manifest, unmistakable स्पष्ट
3. **Policymaker** (noun) – lawmaker, regulator, decision-maker, administrator, official नीति-निर्माता
4. **Shield** (verb) – protect, safeguard, defend, insulate, shelter सुरक्षित करना
5. **Coastal** (adjective) – seaside, littoral, shore-side, maritime, waterfront तटीय
6. **Vagaries** (noun) – fluctuations, whims, caprices, uncertainties, swings अनिश्चितताएँ
7. **Flash flood** (noun) – sudden flood, cloudburst flood, rapid inundation, surge flood, torrent अचानक बाढ़
8. **Landslide** (noun) – mudslide, slope failure, earth slip, debris flow, hillside collapse भूस्खलन
9. **Cause havoc** (phrase) – wreak damage, devastate, wreak havoc, cause mayhem, create chaos तबाही मचाना
10. **Waterlogging** (noun) – flooding, standing water, inundation, water stagnation, soakage जलभराव
11. **Crumbling** (adjective) – dilapidated, decaying, deteriorating, failing, collapsing जर्जर
12. **Infrastructure** (noun) – public works, basic facilities, civic utilities, transport network, built environment आधारभूत संरचना
13. **Disrupt** (verb) – interrupt, derail, disturb, upend, hamper बाधित करना
14. **Congestion** (noun) – traffic jam, gridlock, bottleneck, crowding, clogging जाम
15. **Take a toll** (phrase) – inflict losses, exact a price, cause heavy damage, exact a toll, cost lives नुकसान पहुँचाना
16. **The writing is on the wall** (phrase) – clear warning, obvious sign, ominous signal, clear portent, unmistakable cue इशारा साफ़ है
17. **Decade** (noun) – Period of ten years दशक
18. **Drainage** (noun) – stormwater system, sewerage, outflow network, runoff system, culvert network जलनिकासी
19. **Imperative** (noun) – necessity, must, exigency, mandate, urgent need अनिवार्यता
20. **Anticipate** (verb) – foresee, predict, expect, preempt, envisage पूर्वानुमान करना
21. **Mechanism** (noun) – system, apparatus, framework, protocol, arrangement तंत्र

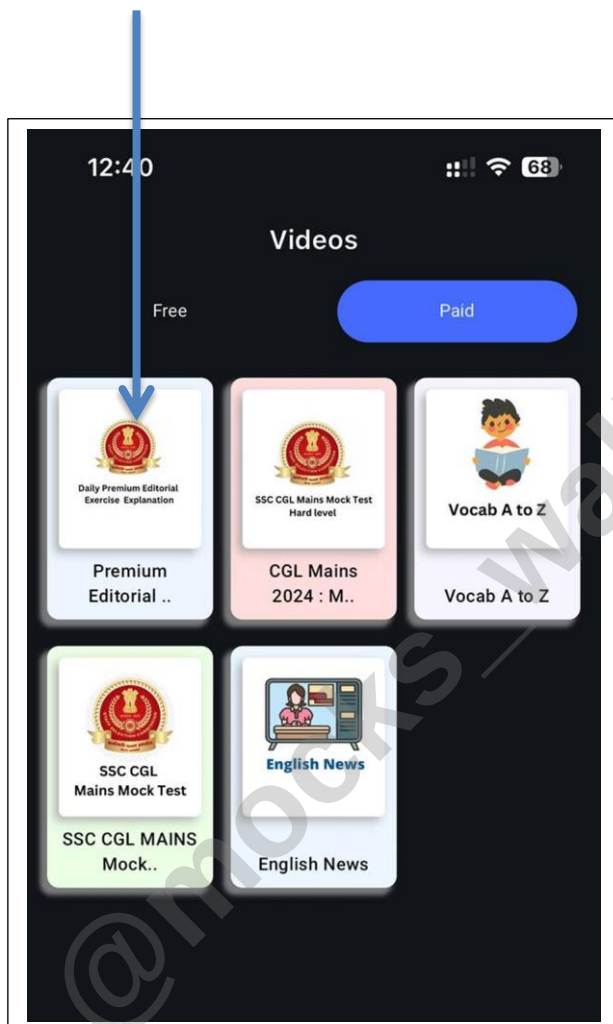
22. **Expose** (verb) – reveal, uncover, lay bare, bring to light, unmask उजागर करना
23. **Frailty** (noun) – weakness, vulnerability, fragility, shortcoming, infirmity कमजोरी
24. **Exacerbate** (verb) – worsen, aggravate, intensify, heighten, compound बढ़ा देना
25. **Idiosyncrasy** (noun) – peculiarity, quirk, distinctiveness, peculiarity, singularity विशिष्टता
26. **Topography** (noun) – terrain, landform, relief, landscape, physiography भू-आकृति
27. **Prone** (to) (adjective) – susceptible to, vulnerable to, liable to, predisposed to, inclined to प्रवृत्त
28. **Global warming** (noun) – climate warming, planetary heating, temperature rise, greenhouse warming, warming trend वैश्विक ऊष्मीकरण
29. **Treacherous** (adjective) – dangerous, perilous, hazardous, deceptive, unreliable खतरनाक
30. **Unleash** (verb) – release, let loose, set off, unleash, unchain शुरू करना
31. **Meteorologist** (noun) – weather scientist, forecaster, climatologist, weather expert, synoptician मौसम-विज्ञानी
32. **Tract** (noun) – area, stretch, expanse, zone, region क्षेत्र
33. **Downpour** (noun) – cloudburst, heavy shower, torrential rain, deluge, pour मूसलाधार वर्षा
34. **Trigger** (verb) – cause, spark, set off, precipitate, activate उत्पन्न करना
35. **Theoretically** (adverb) – in principle, academically, notionally, hypothetically, conceptually सैद्धान्तिक रूप से
36. **Forecast** (verb) – predict, project, foresee, estimate, prognosticate पूर्वानुमान करना
37. **Dense** (adjective) – thick, packed, concentrated, closely spaced, high-resolution सघन
38. **Computing** (adjective) – computational, processing, computer-based, data-processing, high-performance संगणकीय
39. **Reportedly** (adverb) – allegedly, supposedly, by reports, as stated, as per reports बताया जाता है
40. **Step up** (phrasal verb) – scale up, intensify, ramp up, enhance, increase तेज़ी लाना
41. **Enormous** (adjective) – huge, vast, immense, massive, colossal विशाल
42. **Fragile** (adjective) – delicate, vulnerable, brittle, frail, sensitive नाज़ुक
43. **Ecosystem** (noun) – ecological system, environment, biosphere network, habitat system, biotic community पारिस्थितिकी तंत्र

44. **Coordination** (noun) – harmonization,  
alignment, integration, collaboration, concert  
समन्वय

45. **For starters** (phrase) – to begin with, first of  
all, initially, as a first step, at the outset  
शुरुआत के लिए

@mocks\_wallah

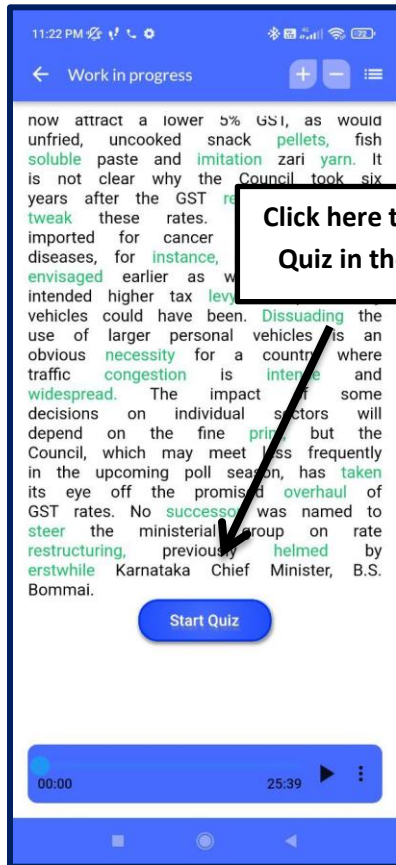
**Video solution of Practice Exercise from this Editorial is exclusively available on App.**



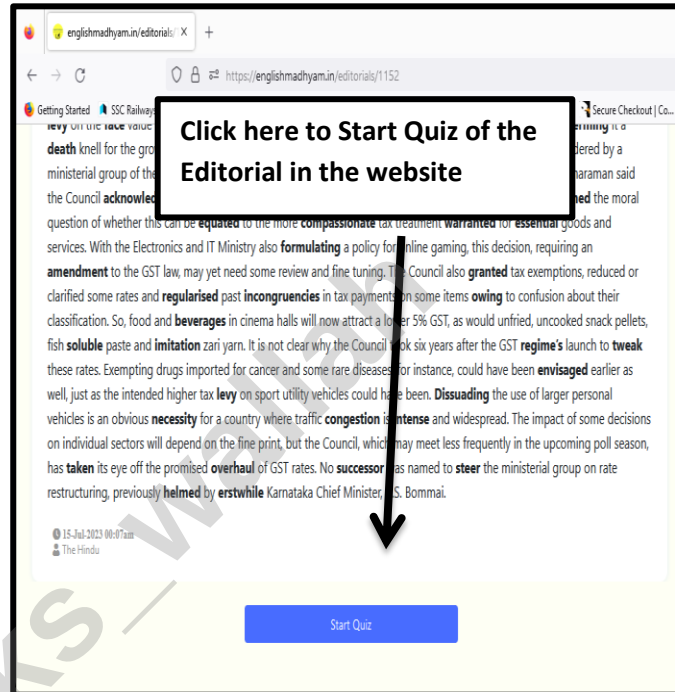
### Summary of the Editorial

1. India faces increasing threats from cloudbursts, flash floods, and landslides across the Northeast, Uttarakhand, Himachal Pradesh, and Jammu & Kashmir.
2. Urban centres like Delhi, Bengaluru, and Mumbai also suffer from waterlogging, traffic chaos, and infrastructural collapse during monsoons.
3. These events have taken a heavy toll on lives, highlighting the urgent need for stronger disaster preparedness.
4. The country has long needed a drainage system upgrade, especially in urban areas.
5. Timely warning systems are crucial to reducing loss of lives and damage from extreme weather events.
6. Accurate data collection and effective communication channels are the foundation of such early warning systems.
7. The recent loss of over 60 lives in J&K exposes weaknesses in India's climate information network.
8. Cloudbursts occur due to a mix of monsoon and cold winds, amplified by Himalayan topography.
9. Global warming appears to intensify these cloudburst events, making them deadlier.
10. A cloudburst is defined as rainfall of 100 mm or more per hour over an area of 30 sq. km.
11. Small areas where cloudbursts occur often lack proper rainfall measuring instruments.
12. Their impacts are not localized — triggering landslides, flash floods, and destruction downstream.
13. Forecasting cloudbursts requires a dense network of weather instruments and advanced computing power.
14. The IMD is working to fill information gaps and better utilize available data.
15. Protecting fragile ecosystems demands coordinated action among IMD, scientists, planners, and all levels of government, including mapping risk zones and regulating construction in vulnerable areas.

## Practice Exercise: SSC Pattern Based



[Click here to download App](#)



[Click here to visit website](#)